

Do Structural Priors Actually Help Unrolled Channel Estimators?

Results digest — Rydberg atomic receivers

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Short version of the unrolled-Transformer study. One experiment family throughout: $N = 32$, $K = 3$, $P = 20$ unless stated, SNR $\sim \mathcal{U}[-10, 20]$ dB, 10 unrolled layers, 1,586,900 parameters, 80k training samples, 2000 paired test realisations. Simulation only.

1. The finding, in one paragraph

We built an unrolled Transformer channel estimator (“URformer”) for the magnitude-only Rydberg problem, added a Hankel low-rank structural prior inside the loop, and measured a clear **+1.209 dB** advantage at high SNR. We believed it. Then we ran a control to characterise where that advantage came from — and the control removed it. Under matched training the same contrast is **+0.078 dB**, a fifteenfold collapse.

The reason is a training convention so common it is rarely stated. The loss is a *per-sample normalised* error averaged over a wide SNR range. Low-SNR samples have far larger normalised error, so they dominate the average: **89.7% of the gradient comes from trials below 5 dB**. The high-SNR regime is therefore underfitted, and *any* mechanism that improves high-SNR behaviour looks like it is adding information. A structural prior is exactly such a mechanism.

The first casualty of this control was our own method. We report it because a result that survives an author’s attempt to keep it is worth more than one that was never tested.

2. Where the gain actually comes from

Before the control, we decomposed the unrolled estimator’s +3.345 dB improvement over the classical EM-GS baseline.

- The learned per-layer **filter** (980 parameters) contributes 0.147 dB.
- The **attention block** (1,585,920 parameters) contributes the remaining 3.198 dB — **96% of the total**.
- The filter’s contribution *falls* with more data ($0.193 \rightarrow 0.147$ dB from 20k to 80k samples), which is what a never-capacity-limited 980-parameter component should do.
- **Unrolling itself does real work**, independently of attention. Against a matched control that applies one Transformer block to a *converged* EM-GS estimate instead of unrolling, the unrolled estimator gains +0.920 dB (CI [+0.888, +0.988]), winning on 86.5% of trials.

So the architecture is doing something real. The question the rest of this digest answers is whether the *structural prior* added on top of it is.

3. The confound

The training loss is the per-sample normalised error $\|\hat{\mathbf{G}} - \mathbf{G}\|_F^2 / \|\mathbf{G}\|_F^2$, averaged over a batch spanning the whole SNR range. A low-SNR realisation has a normalised error orders of magnitude larger than a high-SNR one, so the average is dominated by the low-SNR tail. Measured directly on our own estimator:

- 89.7% of the gradient comes from trials below 5 dB.

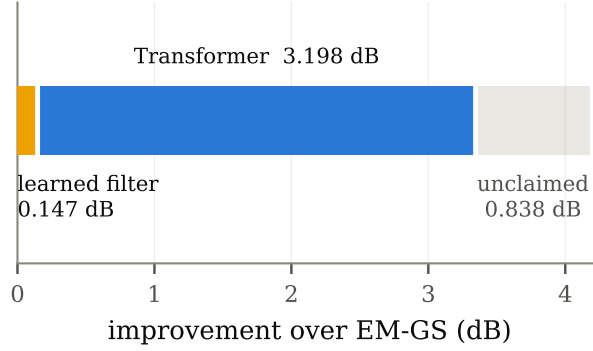


Figure 1: Attribution of the unrolled estimator’s gain over EM-GS at 80k training samples. The attention block accounts for 96% of the +3.345 dB. “Unclaimed” is the remainder of the genie-aided least-squares headroom — a *reference*, not a bound: it presumes the true phase of the noiseless field, which no receiver has.

- The per-bin gradient share spans a factor of 30 across the range (0.465 in the lowest bin against 0.015 in the highest).

The consequence: the high-SNR regime is systematically underfitted, so anything that helps there gets credited with supplying information it did not supply.

4. The control, and what it did to our method

The protocol is cheap, and we recommend running it before any structural prior is credited.

1. **Measure** the per-bin gradient share of the training loss. If it is far from uniform, the regime receiving least gradient is underfitted.
2. **Retrain both arms** — structured and unstructured — with training concentrated on the regime of interest, matching data budget, schedule, seed and initialisation.
3. **Re-measure the contrast.** What survives is attributable to the prior. What does not was compensating for training inadequacy.

Applied to our own estimator over [5, 20] dB:

	Mixed-SNR training	Matched focused training
Structural advantage, $\text{SNR} \geq 5$ dB	+1.209 dB	+0.078 dB (CI [0.050, 0.112])
<i>Absolute NMSE gain from focused training, $\text{SNR} \geq 5$ dB:</i>		
Unstructured arm		+2.653 dB
Structured arm		+1.629 dB

Table 1: A fifteenfold collapse. The mechanism is in the bottom half: the *unstructured* network gains far more from adequate high-SNR training than the structured one does. Given proper gradient in that regime it learns the array geometry by itself, leaving the explicit projection almost nothing to add.

Scope of this negative result. It holds at $N = 32$, $K = 3$, this data budget and the pilot counts tested. It is *not* a general claim that structural priors cannot help unrolled estimators. Two counterweights from the same study:

- The same prior applied to the *classical* estimator — where no training adequacy is at stake — gives a large and stable gain.
- Under a distribution shift in path richness, the structured network degrades *least* (0.48 dB against 0.73 dB).

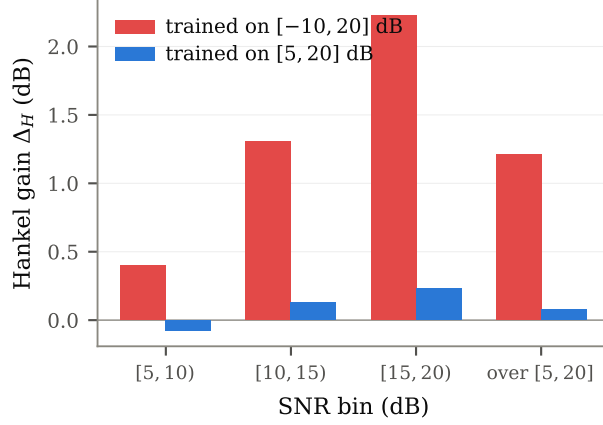


Figure 2: The structural advantage before and after the control. Both arms are retrained identically; only the training SNR range differs. The prior’s apparent value was largely a proxy for the unstructured arm’s high-SNR underfitting.

The prior buys robustness here. It is the in-distribution accuracy claim that does not survive the control.

5. The fix: rebalance the loss

The confound is a property of the loss, so we fixed the loss. Each sample is weighted by a static per-bin factor $w(b) = c/m(b)$, where $m(b)$ is the mean per-sample normalised error in bin b , measured once before training and scaled to unit mean weight. **No schedule, no tuning, no architectural change** — a few lines. It works as designed:

- The per-bin gradient share now spans a factor of 1.9, against 30 before.
- The share below 5 dB falls from 0.897 to 0.427, against an ideal of 0.5 — a slight over-correction.

And the estimation result is **better than we predicted**:

- Against an identically-trained unbalanced baseline, balancing gains +1.902 dB at $\text{SNR} \geq 5$ dB (CI [+1.823, +1.986]), rising monotonically to +2.628 dB in the top bin.
- We had **pre-registered a low-SNR cost** of -0.10 to -0.40 dB, reasoning that the lowest bin’s weight falls about eighteenfold. **There was no cost**: the measured low-SNR change is +0.135 dB (CI [+0.101, +0.172]). Those bins were over-served, not merely well-served.
- The gap to the genie-aided reference at $\text{SNR} \geq 5$ dB falls from +2.897 to +1.000 dB.

Rebalancing the loss recovers considerably more than the structural prior appeared to offer — +1.902 dB against an apparent +1.209 dB — at a fraction of the complexity.

6. A separate trap: pilot efficiency is not pilot-count generalization

A pilot-count curve for a trained estimator can mean two different things, and they are routinely reported as one: **generalization** (one model trained at some P_0 , evaluated at each P) or **efficiency** (a separate model trained at each P). They are not close.

- Trained at $P = 10$, the estimator beats the $P = 20$ model evaluated there by +2.231 dB pooled, with every bin’s CI excluding zero.
- At $P = 35$ the corresponding margin is only +0.635 dB — the model generalizes *upward* in pilot count far better than downward.
- At 5 dB the matched-trained estimator reaches -7.52 dB at $P = 10$, within 0.47 dB of the

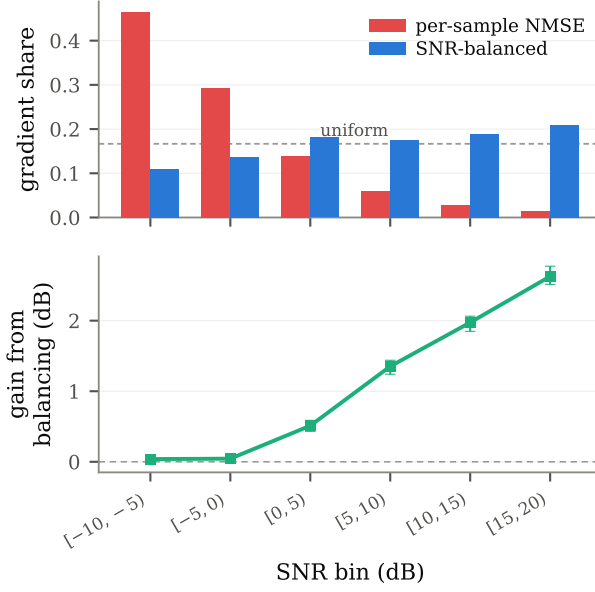


Figure 3: Cause and effect. *Upper:* per-bin gradient share under the per-sample normalised loss and under SNR balancing. *Lower:* the resulting per-bin gain, paired per-trial median with bootstrap 95% CI. Below 5 dB the balanced estimator passes the genie-aided reference by up to 1.99 dB — consistent rather than anomalous, since that reference is restricted to unstructured least squares and an estimator that has learned the channel geometry is not.

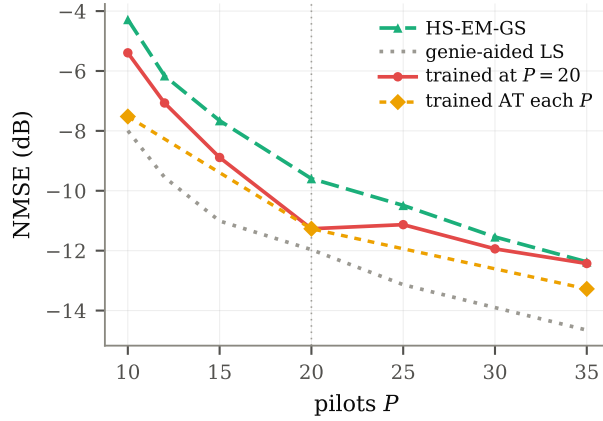


Figure 4: Pilot efficiency against pilot-count generalization at 5 dB, on identical realisations. Dashed diamonds are separate networks trained at each P ; the solid curve is one network trained at $P = 20$. Reported as a single curve, these two quantities are indistinguishable to a reader — and they differ by 2.231 dB.

genie-aided reference, while the $P = 20$ model evaluated there is 2.60 dB short of it.

- The clearest single symptom: the $P = 20$ model is **worse at $P = 25$ (-11.13 dB) than at $P = 20$ (-11.27 dB)**. More measurements, worse estimate — because the curve is tied to its training condition.

7. What this does and does not claim

- **Not a critique of any specific paper.** The convention examined here is near-universal, and our own estimator used it. Our earlier pilot curves had the same ambiguity.
- **Not a reproduction of prior work.** The URformer is implemented faithfully on our geometric ULA model, but the operating point differs and reaching the reported behaviour needed roughly four times the stated data budget. Nothing here validates or refutes a published number.
- **No bound is established.** The reference curve throughout is genie-aided — it presumes the

true phase of the noiseless field — and our own balanced estimator passes it below 5 dB.

- **The negative result is scoped**, to $N = 32$, $K = 3$, one data budget and the pilot counts tested. It is not a claim that structural priors cannot help unrolled estimators anywhere.
- **Simulation only**. No measured hardware, no recorded channel data.

8. What to take away

1. **Run the matched-adequacy control before crediting a structural prior.** Measure the gradient share; retrain both arms on the regime of interest; re-measure. It is cheap and it is decisive.
2. **Check the loss weighting before adding architecture.** A per-bin reweighting of a few lines beat the structural prior it was competing with, at no low-SNR cost.
3. **Say whether a pilot-count curve is efficiency or generalization.** Here the two differ by 2.231 dB.

Full version, with the attribution table, the pre-registrations and the complete limitations:

[paper/sp12/haatim_structural_priors_evaluation.pdf](#). The companion work on the classical estimator (HS-GS) is [paper/master/](#) and [paper/summary/](#). No experiment was rerun to produce this digest.